that some stocks with wide spreads traded better with the $0.05 tick than with the $0.01 tick. The empirical analysis in the Proposing Release sought to identify the thresholds where the TSP tick size change transitioned from harmful, to benign, to beneficial. Specifically, Table 8 provides analysis that examines the impact of the end of the TSP on a wider range of quoted spread profiles than simply tick-constrained or not. This analysis focuses on the end of the TSP, when the tick size was reduced from $0.05 back to $0.01, because that event more closely matches the amendments, which reduce the tick size.

The analysis presented in Table 8 uses a difference-in-difference methodology to study the effect of lowering the tick size from $0.05 to $0.01 on TSP stocks at the end of the TSP.[1267] TSP treated and control stocks are assigned near the end of the TSP into one of four bins ranging from the most tick-constrained in the first bin to the least constrained in the fourth bin.[1268] Key variables such as quoted depth and spreads were measured before and after the tick size was lowered, and difference in difference estimation methods were used to examine how these

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1267 Difference-in-differences is a statistical technique in which the effect that a treatment has on some response variable is estimated by comparing the average change in the response over time in the treatment group to the average change in the control group.

1268 Bin assignments are calculated according to the stock's average quoted spreads for May and June of 2018, near the end of the TSP. Specifically, we use WRDS Intra-day indicators to collect the time-weighted quoted spread for all TSP and control stocks for each trading day in May and June 2018. Then for each stock we calculate the equally-weighted average quoted spread across all trading days. Based on this average, TSP and control stocks are sorted into one of four bins. The first bin is for stocks with quoted spreads ($0.00, $0.06). Empirically, for stocks in the TSP, this bin includes stocks that nearly always traded at the minimum quoting increment of $0.05 during the TSP. The second bin is for stocks with quoted spreads in the range ($0.06, $0.09). For stocks in the TSP, this bin is said to include those stocks with one to two ticks intra-spread during the TSP. The third bin is for stocks that had quoted spreads of ($0.09, $0.15) or approximately 2-3 ticks intra at a $0.05 tick increment. The fourth bin is for stocks with quoted spreads greater than $0.15. The TSP had three test groups: the first group applied the $0.05 tick only to quoting, the second group applied the $0.05 tick to quoting and trading (with exceptions for benchmark and midpoint trades and for certain retail price improvement trades), and the third group applied the $0.05 tick to trades and quotes the same as the second group but also had a trade at rule applied. Barardehi et al., supra note 231 provide similar analysis, and also expand the analysis in many dimensions. Their analysis finds evidence that all key results presented here are robust along many dimensions including the test group analyzed and to many other factors including fixed effects and volatility – factors that one commenter suggested that the Commission should consider in their TSP analysis, see Virtu Letter II at 17.

variables reacted to the tick size change. The analysis uses ordinary least squares[1269] and quantile (median) regressions[1270] to estimate the following regression model:[1271]

$$Y_{j,t} = \alpha_0 + \alpha_P Pilot_j + \alpha_E Event_t + \beta\left(Pilot_j \times Event_t\right) + u_{j,t}$$

where the quantile regression optimizes:[1272]

$$\beta \in \underset{\alpha_0,\alpha_P,\alpha_E,\beta}{\operatorname{argmin}} \sum_{j,t} \rho_{0.5}(u_{j,t})$$

Table 8: Effects of a Reduction in Tick Size on Quoting and Trading Outcomes[a]

	OLS				Quantile (median) regression			
	Quoted spread ($) May & June 2018				Quoted spread ($) May & June 2018			
Spread Bin #	1st	2nd	3rd	4th	1st	2nd	3rd	4th
Depth (100 shares)	-22.5***	-5.30***	-1.55***	-0.51	-11.8***	-3.16***	-0.96***	-0.21***
	[-12.02]	[-7.09]	[-4.40]	[-1.30]	[-16.99]	[-23.52]	[-17.81]	[-4.30]
Depth ($1,000)	-16.7***	-8.41***	-4.67***	-2.06***	-11.2***	-7.27***	-3.96***	-1.48***
	[-14.58]	[-10.94]	[-7.82]	[-3.66]	[-22.04]	[-20.70]	[-12.58]	[-4.14]
Quoted Spread ($)	-0.033***	-0.027***	0.023***	0.12***	-0.034***	-0.031***	0.012**	0.12***
	[-18.71]	[-6.46]	[2.99]	[5.51]	[-35.41]	[-10.31]	[2.03]	[6.80]
	-0.0049***	-0.00097*	0.00034	0.0046***	-0.0041***	-0.0014***	0.00021	0.0034***

1269 Ordinary least squares (OLS) regression refers to a statistical technique for estimating the linear relationship between an independent variable and dependent variables by minimizing the sum of squared errors between the estimate and the observed independent variable. The use of OLS and quantile regressions is common in the literature on the TSP pilot.

1270 The primary advantage to quantile regressions is that they are less sensitive to outliers that can affect mean inference in OLS. Thus, median regressions provide additional robustness to the analysis and ensure that results are not driven by outliers.

1271 In this equation the variable *Y* denotes the response variable of interest such as quoted spread and depth. The subscripts *j* and *t* serve to index stocks and days respectively. α_0, α_p, α_e, and β are coefficients (to be estimated), and $u_{j,t}$ is the error term. $Pilot_j$ is an indicator variable that equals 1 if stock *j* was in the treatment group, or 0 if stock *j* was in the control group. $Event_t$ is an indicator variable which is equal to 1 if the day *t* was post the treatment event and equals 0 otherwise. Table 8 reports the difference-in-difference estimator of β for a different response variable *Y* across the different quoted spread bins. One commenter criticized this model for failing to include fixed effects and not controlling for other criteria such as volatility See Virtu Letter II at 15. A very similar analysis, which did consider fixed effects and a host of control variables including volatility, is included in Barardehi et al., supra note 231. Their analysis showed that all key results were economically unchanged when considering fixed effects and a host of control variables.

1272 In this equation $u_{j,t}$ is the error term from the previous regression specification equation, supra note 1271, and the loss function is defined as: $\rho_\tau(u) = \tau \max(u,0) + (1-\tau) \max(-u,0)$; where $0 < \tau < 1$.

Relative quoted Spread	[-9.59]	[-1.80]	[0.53]	[3.30]	[-8.54]	[-6.89]	[0.74]	[4.66]
Effective spread ($)	-0.027***	-0.026	0.029***	0.038**	-0.026***	-0.021***	-0.0018	0.051***
	[-4.97]	[-1.43]	[5.17]	[2.16]	[-58.10]	[-12.81]	[-0.63]	[4.81]
Relative eff. spread	-0.0039***	0.00043	0.0055	0.0028***	-0.0030***	-0.0010***	-0.00013	0.0016***
	[-3.12]	[0.17]	[1.36]	[4.42]	[-10.78]	[-9.58]	[-1.09]	[3.23]
Cancel-to-trade	5.10***	6.69***	7.56***	18.8***	4.56***	5.49***	6.87***	12.3***
	[5.99]	[6.38]	[6.79]	[8.44]	[7.75]	[7.79]	[10.44]	[10.61]
Odd-lot rate (%)	4.89***	5.61***	2.85***	1.49**	5.59***	6.39***	3.29***	1.85**
	[9.62]	[8.04]	[4.35]	[2.15]	[8.02]	[8.99]	[4.72]	[2.51]
Realized spread ($)	-0.014***	-.0099***	.00037	0.040***	-0.014***	-0.013***	-0.0068***	0.038***
	[-27.94]	[-7.43]	[0.12]	[4.45]	[-48.36]	[-17.96]	[-5.13]	[5.64]
Relative real. spread	-.0024***	-.00032	-.00039	.0014**	-.0014***	-.00054***	-.00013***	.0012***
	[-11.82]	[-1.36]	[-1.25]	[2.37]	[-14.08]	[-12.52]	[-2.77]	[3.65]
Volume (1,000 shares)	26.5	30.3**	12.5	-5.41	19.1	3.35	0.20	-3.25**
	[1.30]	[2.13]	[1.32]	[-1.07]	[1.42]	[0.40]	[0.04]	[-2.44]
Cum Depth 10c from mdpt	-0.17***	-0.26***	-0.27**	-0.34**	-0.49***	-0.54***	-0.45***	-0.63**
	[-3.93]	[-5.00]	[-2.59]	[-2.37]	[-5.51]	[-6.29]	[-4.91]	[-3.15]
Cum Depth -10c from mdpt	-0.22***	-0.19***	-0.37***	-0.45**	-0.49***	-0.42***	-0.50***	-0.79**
	[-5.28]	[-3.74]	[-3.44]	[-2.83]	[-6.33]	[-5.21]	[-5.68]	[-2.75]
CRT 10 round lots	-0.026***	-0.001	0.035***	0.14***	-0.037***	0.085***	0.035***	0.075**
	[-19.56]	[-0.19]	[3.99]	[1.03]	[-2.72]	[2.75]	[4.20]	[2.37]

[a] This table presents the effects of a reduction in minimum tick size from $0.05 to $0.01 cent on various quoting and trading outcome variables. The first bin is for stocks with quoted spreads ($0.00, $0.06). The second bin is for stocks with quoted spreads in the range ($0.06, $0.09). The third bin is for stocks that had quoted spreads of ($0.09, $0.15). The fourth bin is for stocks with quoted spreads greater than $0.15. A difference in difference regression with no control variables is estimated using data covering Control, Test Group 2, and Test Group 3 TSP stocks from 08/01/2018 – 11/30/2018. All observations are at the stock day level. For each outcome variable Y_{jt}, listed in the left-hand side column, the table presents only the difference in difference coefficient estimates that indicate the effect of the TSP on the dependent variable. Estimates are performed by past quoted spread subsamples that decompose the sample based on average quoted spreads during May -June of 2018. Among the outcomes' variables, the quoted spread refers to the distance between the NBBO midpoint and the NBBO quote. The effective spread is the distance between the NBBO midpoint and the realized trade price; the realized spread is the distance between a future NBBO midpoint (5-minutes ahead) and the trade price. Relative spread measures are calculated as the spread scaled by the NBBO midpoint. The cancel-to-trade ratio is the daily number of order cancellations divided by the number of trades, for displayed orders. The odd-lot rate is the percentage of trades in a day which executed against an odd-lot quote. CRT 10, or the cost of a round-trip trade of 10 round lots, measures the cumulative transaction costs from buying and then immediately selling 10 round lots. The CRT assumes that an order that is larger than the displayed depth at the best price will not execute in full at that price. Instead, the assumed unfilled portion will execute at worse prices until completely filled with displayed depth. All data are Winsorized at the 1% and 99% level. The numbers in the [] brackets reflect t-statistics that are based on two-way stock-and-date

clustered standard errors. Symbols *, **, and *** reflect statistical significance at 10%, 5%, and 1% type-1 error levels.

As discussed in the Proposing Release, this analysis provides evidence of a fundamental tradeoff between accurate pricing on one hand and incentives for liquidity provision on the other. Across all specifications, the end of the TSP was associated with a decrease in depth at the NBBO, when the tick size was reduced from $0.05 to $0.01, as signified by the negative and, in most cases, statistically significant coefficients reported. The reduction in shares available at the NBBO was the greatest for stocks with tighter quoted spreads and smaller for stocks with wider quoted spreads. The finding that tighter quoted spread stocks experience the greatest decline in depth at the NBBO is consistent with the idea that, for these stocks, the $0.05 tick was the most constraining, and so liquidity that would have naturally spread out within the quoted spread given a smaller tick, bunched at the wider tick increments, and that once the tick-constraint was relaxed this liquidity naturally spread out over the additional price levels. For less tick-constrained stocks, the bunching was less severe since liquidity already had some room to spread out.

One commenter stated that the Commission did not provide any analysis of the effect of the proposal on displayed liquidity and liquidity deeper in the book, including with respect to less liquid securities and during times of market stress.[1273] The Proposing Release did examine the effect of a tick reduction on displayed liquidity and cited academic literature for further analysis.[1274] The same commenter stated that reducing the tick as presented in the proposal

[1273] See Citadel Letter I at 9.

[1274] See Proposing Release, supra note 11, section V.D.1.

would reduce depth at the NBBO by more than 82%.[1275] The Commission acknowledges that, given the magnitude of the reduction in the proposal, it is conceivable that such a reduction could have occurred for some stocks. Barardehi et al. (2022) document that depth at the NBBO was 50% lower with the $0.01 tick compared to the $0.05 tick. However, this was only true for tick-constrained TSP stocks. For stocks with wide quoted spreads, depth was only about 16% lower with the smaller tick size. The TSP was associated with a 1:5 tick size split, and the proposal that the commenter was commenting on would have created a 1:10 split for some stocks relative to the baseline. In contrast, the adopted amendments create a smaller split than either a 1:5 or a 1:10 split. The Commission does expect depth at the NBBO to decrease for stocks receiving the smaller tick size with the TSP analysis providing a likely higher end estimate of the magnitude of the decrease since the TSP was a bigger change to the baseline than the adopted amendments.

For stocks in the first or second bins, Table 8 demonstrates that lowering the tick to $0.01 leads to significantly lower quoted spreads. These stocks went from having approximately 1-2 ticks inside the quoted spread, with a $0.05 tick, to having 1-10 ticks inside the quoted spread, with a $0.01 tick. This finding is consistent with the idea that for stocks that are tick-constrained the effect of decreasing the tick size will narrow quoted spreads by improving competition. For the stocks in the third and fourth bins, the story is different, as the reduction in the tick size leads to wider quoted spreads. These stocks went from having more than two or more ticks within the quoted spread, with a $0.05 tick, to having more than 10 ticks within the quoted spread, with a $0.01 tick. This result is consistent with the idea that for wider quoted spread stocks, the prevailing effect of reducing the tick size was to increase transaction costs and widen spreads by

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[1275] See Citadel Letter I at 5.

fragmenting liquidity and increasing the risk of pennying which made trading more costly leading to wider quoted spreads. This pattern of results – namely narrower spreads for the first and second bins and wider spreads for the fourth -- holds regardless of whether dollar spreads, relative spreads, OLS, or quantile regressions are used, suggesting this is a robust outcome of the end of the TSP.

The pattern for effective spreads is similar to that observed for quoted spreads. Effective spreads measure the average realized transaction cost for trades as it measures the absolute distance between the realized trade price and the NBBO midpoint at the time of the trade. Effective spreads do not always equal quoted spreads because trades can execute inside the NBBO for numerous reasons, such as odd-lot trades, midpoint trades, and hidden orders. For stocks in bin one – i.e., stocks for which the $0.05 tick was the most restrictive – all specifications suggest that reducing the tick size was associated with a decrease in realized transaction costs as measured by effective spreads. For stocks in bin four, those with the widest quoted spreads prior to the tick size reduction, all specifications suggest that the reduction in the tick size leads to an increase in transaction costs, measured by effective spreads. For stocks in between these extremes in bins two and three, the results are not as uniform. For stocks in bin two, the sign of the coefficients for all estimates (dollar effective spreads, relative effective spreads, OLS, and quartile regressions) suggests lowering the tick size decreased effective spreads, although not all specifications agree as to statistical significance. The OLS regressions suggest that the effect was statistically insignificant, while the quantile regressions found a statistically significant effect and suggest that effective spreads decreased. For stocks in the third bin, the analysis did not find a consistent, statistically significant change in effective spreads, or in other words, moving from roughly two to three ticks within the spread to ten to fifteen ticks did not appear to reliably help or harm transaction costs as measured by effective spreads.

These results, like the results for quoted spread, suggest that for stocks for which the narrowing of the tick size meant that the stock went from having less than 2 ticks within the quoted spread to 1-10 ticks within the quoted spread, the effect of reducing the tick was beneficial in terms of reducing transaction costs. For stocks with very wide quoted spreads, reducing the tick size appeared to harm liquidity, which is consistent with fragmentation and pennying being the prevailing effect.

The theoretical discussion above suggests that executing a larger order may become more complex with a smaller tick size – meaning it may take visiting more venues as well as executing across more price levels to execute an order with a smaller tick size. This potential outcome is explored using the "cancel-to-trade" ratio. A higher ratio indicates more frequent canceling of orders per the amount of trading volume, and it is an indication that market participants are more active in managing their quotes and their order strategies. In this analysis, both the OLS and the quantile regressions confirm that a smaller tick resulted in a statistically significant increase in the cancel-to-trade ratio, suggesting more complexity. Additionally, the magnitude of the effect is increasing in the quoted spread, with wider quoted spreads having larger coefficients, suggesting a larger effect in the cancel-to-trade ratio for stocks with wider spreads. This pattern is consistent with pennying and increased complexity having a greater impact on stocks with wider quoted spreads. These stocks are unlikely to receive the smaller tick size under the adopted amendments.

The analysis also looks at the effect of lowering the tick size at the end of the TSP on the usage of odd-lot orders. Across all quoted spread bins, the usage of odd-lot orders increases when the tick size decreases. This finding is consistent with the notion that liquidity will be spread out over more levels leading to an increased use of odd-lot orders to allow liquidity providers to offer smaller levels of liquidity at finer price increments.[1276] This result also suggests that a lower tick size increases the need for market participants to have ready access to odd-lot information given that the lower tick size can be expected to increase the usage of odd-lot quotes.

Effective spreads provide a measure of liquidity providers' revenue for the immediate execution of an incoming order and the contrasting economic effects also have implications for how liquidity providers' revenue will be affected by a lower tick. The effective spread captures the liquidity premium, paid by those submitting orders for immediate execution, and can theoretically be decomposed into two components: *Effective Spread = Realized Spread + Price Impact*.[1277] One component of the effective spread is the price impact or adverse selection component. It is the change in the NBBO midpoint at the time of trade to some point in the future. This component of the effective spread captures the portion of the effective spread liquidity providers lose from trading with investors who are more informed than they are and is also referred to as the adverse selection component of the bid-ask spread. The remainder of the effective spread, after removing the adverse selection component, is the realized spread. This

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1276 See also Virtu Letter II at 6.

1277 Effective spreads can be interpreted as what liquidity providers expect to earn from providing liquidity, assuming that prices do not change before the liquidity provider is able to unwind its position and realize its profit. Under this interpretation, realized spreads would proxy for what liquidity providers actually earn, taking into account that the market price may have moved against the liquidity provider before it could unwind its position. *Effective Spread = Realized Spread + Price Impact.* For a full mathematical decomposition of effective spreads into realized spread and price impact components see Peter N. Dixon, Why Do Short Selling Bans Increase Adverse Selection and Decrease Price Efficiency, 11 REV. ASSET PRICING STUD. 122 (2021) app. at 165.

portion of the effective spread acts as a proxy[1278] for the compensation to the liquidity provider for its non-adverse selection costs. If a smaller tick decreases revenue for liquidity providers, by allowing bid and ask prices to more accurately reflect supply and demand, then this effect should manifest as a decrease in realized spreads for liquidity providers. However, if increased order book fragmentation and pennying risk increase the cost of providing liquidity, then liquidity providers will need to be compensated for these costs in order to provide liquidity and, thus, realized spreads will increase. To the extent that the two effects offset one another, realized spreads might not change.

For tick-constrained stocks in bin one, the analysis indicates a decrease in realized spreads across all specifications, and when using dollar or relative realized spreads when the tick size was reduced from \$0.05 to \$0.01. This result is consistent with the notion that liquidity providers' non-adverse selection revenues will decrease due to bid and ask prices being more reflective of supply and demand with a smaller tick. The opposite occurs for stocks with wide quoted spreads in bin four, where realized spreads increase significantly – consistent with liquidity providers needing to be compensated for the increased cost and complexity associated with trading a wide quoted spread stock in a small tick environment. For stocks in the middle two bins, the effect of lowering the tick size on realized spreads is unclear, as about half of the

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[1278] Realized spreads do not measure the actual trading profits that market makers earn from supplying liquidity. In order to estimate the trading profits that market makers earn, we would need to know at what times and prices the market maker executed the off-setting position for a trade in which it supplied liquidity (e.g., the price at which the market maker later sold shares that it bought when it was supplying liquidity). If market makers offset their positions at a price and time that is different from the NBBO midpoint at the time lag used to compute the realized spread measure (Rule 605 realized spread statistics are measured against the NBBO midpoint 5 minutes after the execution takes place), then the realized spread measure is an imprecise proxy for the profits market makers earn supplying liquidity.

specifications indicate no change in realized spreads while the other half indicate lower effective spreads.

One commenter states that the Proposing Release did not address how the rule would affect institutional transaction costs, given that institutional traders frequently trade, and are concerned with sourcing larger quantities.[1279] However, as considered here, and in the Proposing Release, the Commission considered multiple measures of depth beyond the NBBO, and Barardehi et al. (2022) also considers more. This depth beyond the top of the book analysis uses MIDAS data to study how the tick size change affected liquidity deeper in the book. Analyzing liquidity deeper in the book is valuable because it gives an indication of how trading larger orders, which must go deeper in the book to be fulfilled may be affected by a change in the tick size. Specifically this analysis calculates the daily average cumulative shares available at $0.10 above and below the midpoint for control and treated stocks, and uses the same difference-in-difference analysis to examine the effect of reducing the tick size on cumulative depth.[1280] Our analysis suggests that reducing the tick size also reduced the total depth available deeper in the book with the coefficient for bin 4 – i.e., those with the widest quoted spreads – being the largest in magnitude. This finding is consistent with a smaller tick discouraging the posting of displayed liquidity due to pennying concerns for stocks with wide quoted spreads.

These depth of book findings do not directly imply that trading deeper in the book became more expensive for two reasons. First, research suggests the use of non-displayed

1279 See MEMX Letter at 10.

1280 In the regressions we take the natural log of shares available. This conversion helps standardize shares available for stocks with different prices by making the interpretation in terms of percentage changes. See also e.g., STA Letter at 6 and CCMR Letter at 24 suggesting that a smaller tick size could affect depth deeper in the book.